

Neuro4Chess: A Neuro-First Hierarchical Reward Framework for Multi Agent Decision Making in 4 Player Chess

ISRC-CN3 Summer School 2026

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Contents

1. Motivation
2. Before We start: Why Neuro?
3. The Neuro-First Framework: Mathematical Foundations and Formulations of the NeuroEngine
4. Clear discussion of benchmarks and metrics comparing the CNNEngine and NeuroEngine
5. The Comprehensive Picture: How myriad modules combine to produce 'intelligent' decision making
6. Results
7. Limitations
8. Further Directions (Very exciting, stay tuned!)

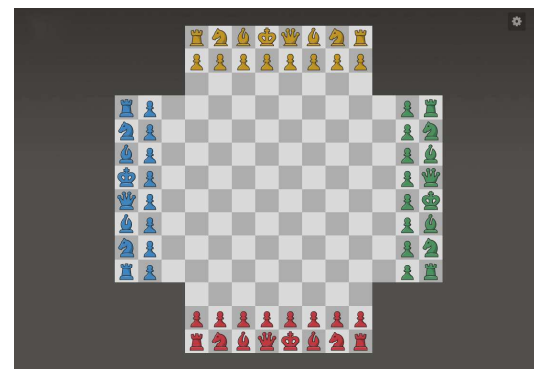
1. Motivation

- How did the idea originate?
- Addressing a critique: Not another chess engine!
- Characteristics of 4 Player Chess as a testbed

Overarching Aim:-

- a) Converge to strategies that promote optimal behaviour.*
- b) What is the mathematical nature of that behaviour?*
‘Building a comprehensive framework and defining metrics to analyse and interpret that behaviour.’
- c) Perform better than a ‘black-box’ agent benchmark*

- Given this broad goal, what is the right path ahead?



Chess.com's 4 player chess variant, which served as the backbone for encoding the fixed rules of the current implementation

2. Before we start: Why choose Neuro?

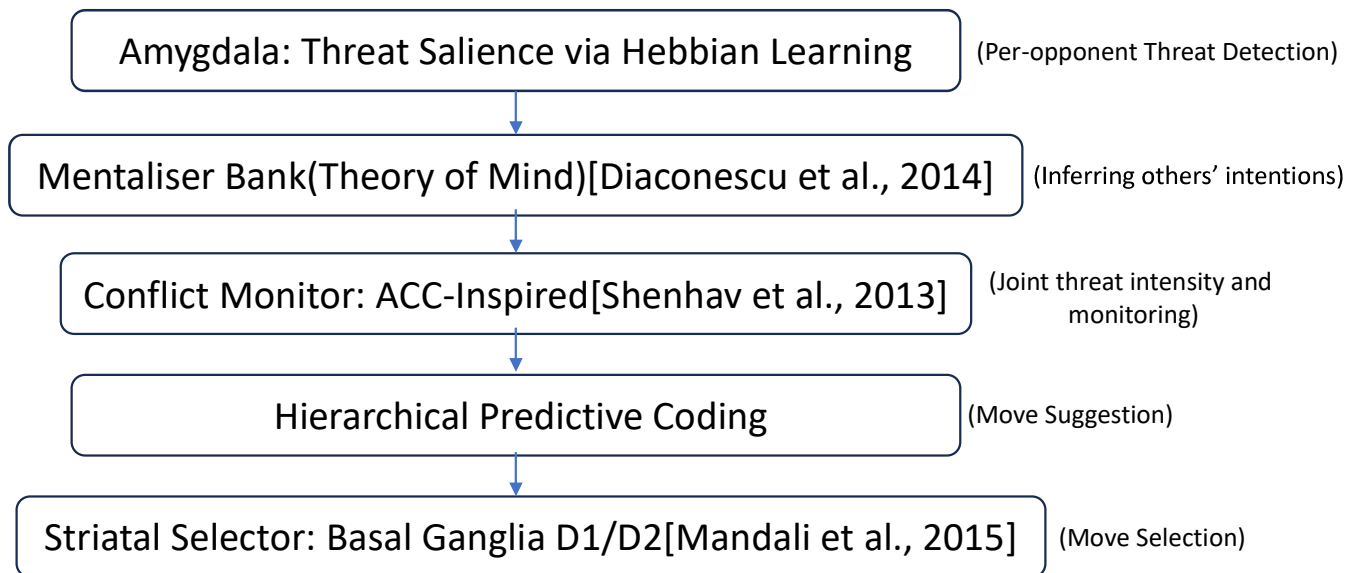
- The only known **self-credible** form of intelligence that has successfully evolved neural mechanisms to perform this exact task.
- Biological brains had realized and implemented multi-agent decision making long before the very genesis of this socio-intellectual challenge was explicitly modelled into strategy and board games!
- Max^n, paranoid search and other paradigms designed to solve multi-player games: The problems of causality and dependent states.



Michael Polanyi, who formulated Polanyi's Paradox: "We know more than we can tell." With every new innovation, we inadvertently try to draw parallels to the brain, but at least to me, the reverse: tackling exciting challenges using the brain's principles as a source of inspiration seems more promising

3. The Neuro-First Framework: Mathematical Foundations and Formulations of the NeuroEngine

The system comprises of five modules, each derived from an identified neural circuit and operating in a fixed processing hierarchy.



4. Clear discussion of benchmarks and metrics comparing the CNNEngine and NeuroEngine

To construct the CNNEngine, we initialize the neural net during evaluation with the following:-

$$E = (\alpha \cdot H + (1 - \alpha) \cdot E_{net}) \times 100$$

where H refers to the handcoded eval function and E_{net} refers to the evaluation net. This acts as a buffer for the CNN engine until it runs through enough training rounds. Initially, α is set very high (around 0.9) and gradually reduced with improvement in play. **(Curriculum Learning)**

Metrics used in hand-coded evaluation function: (Normalized to 1)

1. King Safety: 0.3 | 2. Material: 0.2 | 3. Queen count: 0.1 | 4. Bishop Pair: 0.05 |
5. Passed Pawn: 0.08 | 6. Center Control: 0.07 | 7. Mobility: 0.1 | 8. Third Party Alliance: 0.1

Third party alliance refers to temporary alliances formed with opponent teams to defeat another opponent. *Tertius Gaudens!*

5. The Comprehensive Picture: How myriad modules combine to produce 'intelligent' decision making

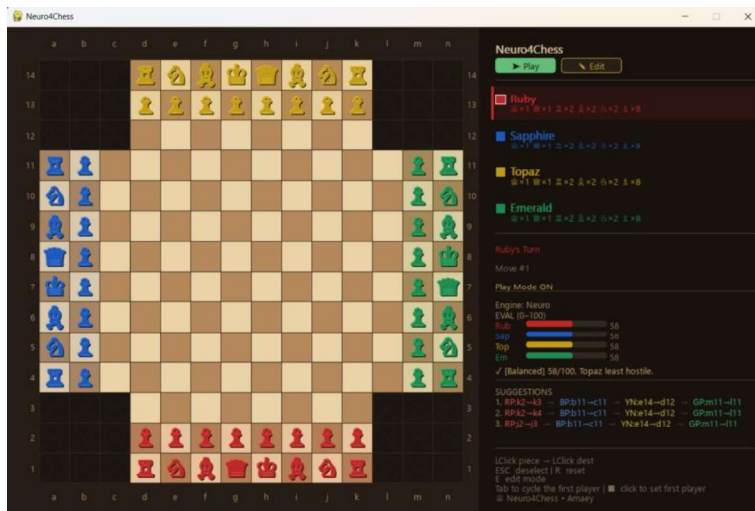
Programmer's Log: The whole project has a codebase of 6 python files, namely:-

(Walkthrough)

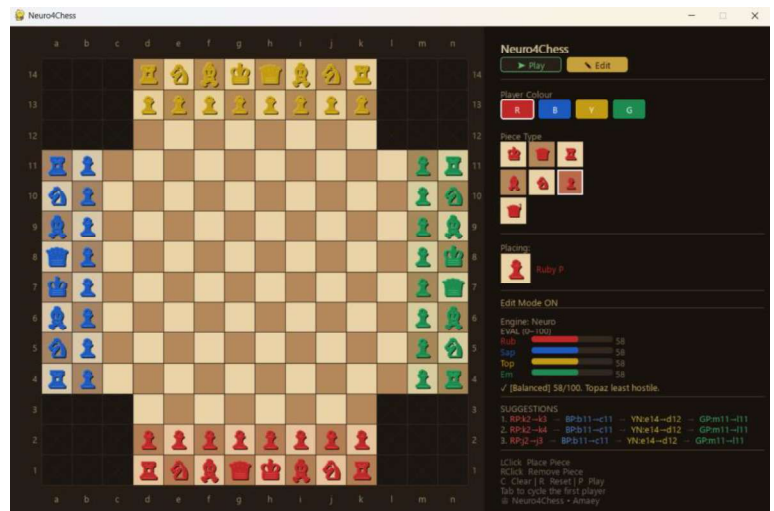
1. `move_generator.py`: Houses all fixed rules of 4 player chess, such as castling, en passant etc.
2. `move_logger.py`: Houses all records of previous moves and previously played games as well
3. `CNN_Engine.py`: Houses the modules of the CNN engine.
4. `Neuro_Engine.py`: Houses all the modules of the Neuro engine.
5. `Training.py`: The training ground where both engines duke it out! Every training run has 'x' `Neuro_Engines` v/s '4-x' `CNN_Engines`, where $x \in [0,4]$. Depends on how much a particular Engine needs to improve.
6. `main.py`: The final pitstop where every physical aspect of the 4 player chess board and other functionalities within pygame is encoded.

(Run `main.py` and see the results! More details in the next page.)

6. Results



Play Mode



Edit Mode

Demo coming up!

7. Limitations

- a. **Overlap between modules**: While overlapping of brain functions between different parts of the brain is biologically grounded, in the code, it becomes a little harder to discern which modules perform a specific problem which needs to be debugged. The goal is to keep the implementation open and understandable as possible. Looking forward to any ideas as to how to find the balance between *neuro-faithfulness* and *modularity in code*.
- b. **Results are extremely preliminary**: Stronger results will need better benchmarks, especially with RL frameworks which I am still getting familiar with. Would be grateful for any open source model suggestions where I can consider benchmarking at scale, like Brain-Score but extended beyond vision and language.
- c. **Programming Language**: The current model is completely written in Python(using pygame) as a personal preference, whereas most traditional chess engines have historically used C++ for better search spaces and execution speed.

8. Further Directions

4 Player Chess is just the starting point! It is a complete information (partial cognitive information) game and hence relies solely on strategy. Some partial information multi-player games that are genuinely harder to solve include:-

1. **Mahjong**
2. **Chinese Checkers**
3. **Chaukabara**: An ancient Indian game, widely considered as a precursor to Ludo

And **Catan**, which is probably the final frontier for strategy games in general.

Beyond these, I'm sure dynamic coalitions and conflicts will be a part of a lot of interesting **embodied AI** and **world model** research, and I can't wait to see where we move forward as a community in the next few years!

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Hope you enjoyed the talk! Now open for QnA.

Zenodo DOI
(Abstract only)

